4. NETWORK THREATS, MITIGATION AND PHYSICAL SECURITY

Course: Diploma in Cyber Security

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Introduction

Network security is not only about software. Physical protection of computers, servers and network devices is also important. Common threats include phishing, DDoS, man-in-the-middle attacks, malware and password attacks.

Task 1: Common network threats

The table below gives simple real-world examples and one basic protection idea.

Threat Name	Real-World Example	Basic Protection Idea
Phishing	An employee receives an urgent fake email pretending to be IT support asking to reset their password via a link.	Use multi-factor authentication (MFA) and train users to recognize suspicious emails.
Malware / Ransomware	A user downloads a malicious email attachment that encrypts files on the network and demands a ransom.	Keep antivirus/EDR software updated, patch systems, and maintain regular offline backups.
Man-in-the-Middle (MitM)	An attacker on public Wi-Fi intercepts unencrypted web	Always enforce HTTPS encryption (TLS) and use a trusted

Threat Name	Real-World Example	Basic Protection Idea
	traffic between a laptop and a website.	VPN on public networks.
Denial of Service (DoS / DDoS)	A botnet floods a company's web server with millions of requests, causing the website to crash.	Use rate limiting, firewalls, and cloud-based DDoS protection services (e.g., Cloudflare).
Brute Force Attack	An automated script attempts thousands of password combinations on an open login or remote desktop port.	Enforce strong password complexity, account lockout policies, and disable direct public exposure of ports.

Task 2: Brute-force PIN simulation

For a classroom demonstration, a script can check every 4-digit combination against a preset lab PIN and count the attempts. This should be run only against the preset value in the script, not against a real account or service.

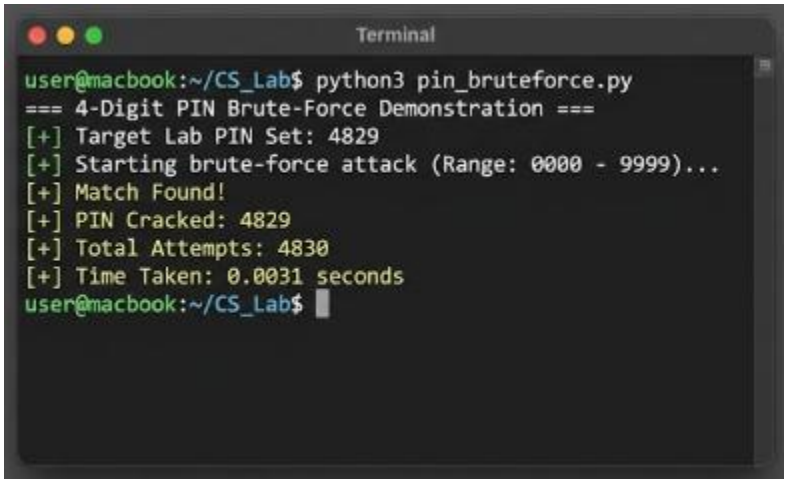


Figure: Execution Output of Classroom 4-Digit PIN Brute-Force Demonstration

Task 3: Physical security risks

Server rooms can face unauthorised entry, theft, overheating, water damage and power problems. Access control, CCTV, locks, temperature monitoring and backup power are common controls.

Physical Risk / Threat	Potential Impact	Recommended Security Control / Mitigation

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Unauthorized Entry	Data theft, hardware tampering, insertion of rogue devices (e.g., keyloggers, packet sniffers).	Biometric scanners, RFID badge access control, mantrap doors, and multi-factor physical authentication.
Hardware Theft	Loss of proprietary data, exposure of storage drives, direct operational disruption.	Heavy-duty physical server rack locks, security cages, and 24/7 CCTV surveillance with retention logging.
Overheating / HVAC Failure	Thermal throttling, premature component failure, abrupt	Redundant HVAC (heating, ventilation, and air conditioning) units, cold/hot aisle

Physical Risk / Threat	Potential Impact	Recommended Security Control / Mitigation
	system shutdowns leading to data corruption.	containment, and automated IoT temperature/humidity sensors with SMS alerts.
Water Damage & Flooding	Short circuits, permanent motherboard/PSU destruction, complete facility outages.	Water leak detection ropes under raised floors, installing server rooms on upper floors (avoiding basements), and clean-agent gaseous fire suppression (FM-200 / Novec 1230) instead of water sprinklers.
Power Outages &	Unscheduled downtime,	Industrial Uninterruptible

Physical Risk / Threat	Potential Impact	Recommended Security Control / Mitigation
Voltage Spikes	corrupted file systems, damage to sensitive microchips.	Power Supplies (UPS) for instant battery fallback, surge protectors, and dedicated on-site diesel backup generators.

Task 4: Office floor plan

Draw a small office and mark at least three controls, such as CCTV near the entrance, biometric access for the server room and cable locks for network equipment.

Figure: Secure Office Physical Layout and Security Controls

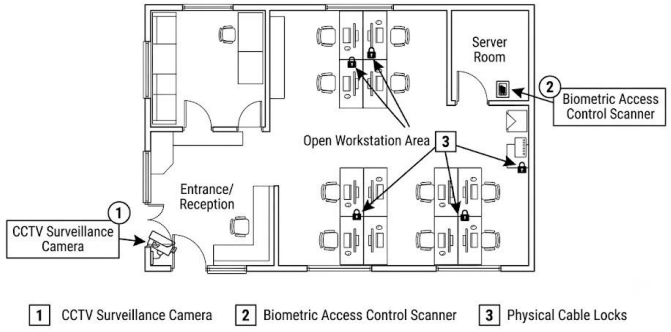


Figure: Office Layout with Integrated Physical Security Controls

Task 5: Risk mitigation checklist

A college lab checklist can include updating systems, using strong passwords, enabling firewall protection, checking suspicious links, restricting server-room access, keeping backups, and monitoring unusual network activity.

Threat	Simple Example	Basic Mitigation
Phishing	Fake login link sent by message	Check the URL and use MFA
DDoS	Service receives huge amounts of traffic	Use filtering and DDoS protection
Man-in-the-middle	Attacker tries to observe traffic on unsafe Wi-Fi	Use HTTPS and trusted networks
Malware	Malicious file infects a PC	Use security software and updates
Password attack	Weak password is guessed	Use long unique passwords and MFA

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.